

Confidence-Gated Adaptive Back-off

Generalizing ReAct's hybrid trigger beyond step-exhaustion

Assignment 3 — Paper Improvement Exploration

Eugene · ReAct (Yao et al., ICLR 2023)

the paper's own fallback rule doesn't fire

Roadmap

4 acts

- **Reproduce** — ReAct/CoT/Act/Standard + 2 hybrids, `z-ai/glm-5.2`, n=100/domain
- **Diagnose** — read the failure traces, find what the paper's trigger misses
- **Fix** — generalize the trigger into 3 validated signals ("CGA")
- **Prove** — validate signals *before* paying for the ablation, then ablate

HotpotQA · multi-hop QA FEVER · claim verification

What ReAct actually does

Thought → Act → Observe

- Interleaves **reasoning** (*Thought*) with **actions** (*Search, Lookup, Finish*) against a live Wikipedia environment
- Section 3.3 adds two **hybrid** fallbacks with chain-of-thought self-consistency:
 - **ReAct** → **CoT-SC** — back off to an n -sample majority vote when ReAct fails
 - **CoT-SC** → **ReAct** — back off the other direction when the vote is too split
- Both directions need a **trigger**: a rule for *when* to switch strategies

The reproduction

A2 → A3

- Reimplemented WikiEnv, all 4 base strategies + both hybrids
- Model: `z-ai/glm-5.2` (OpenRouter) standing in for PaLM-540B
- Assignment 2: **n=10** gate only — too small, trends didn't make sense
- Assignment 3: scaled to **n=100 per domain**, same items, same seed

7methods

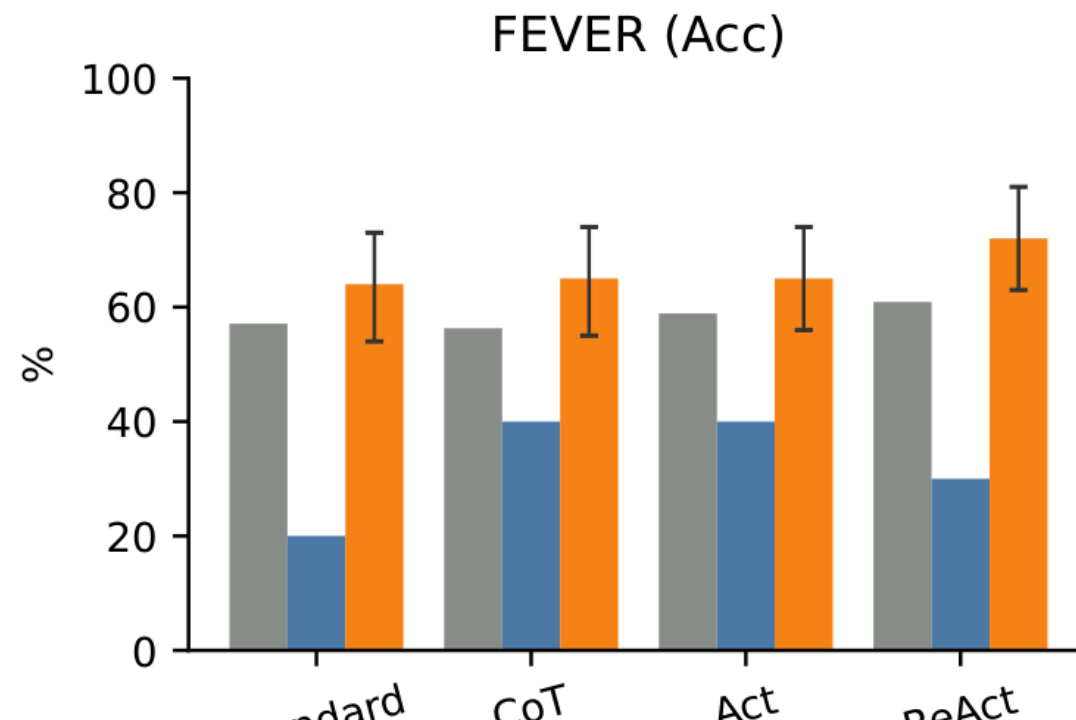
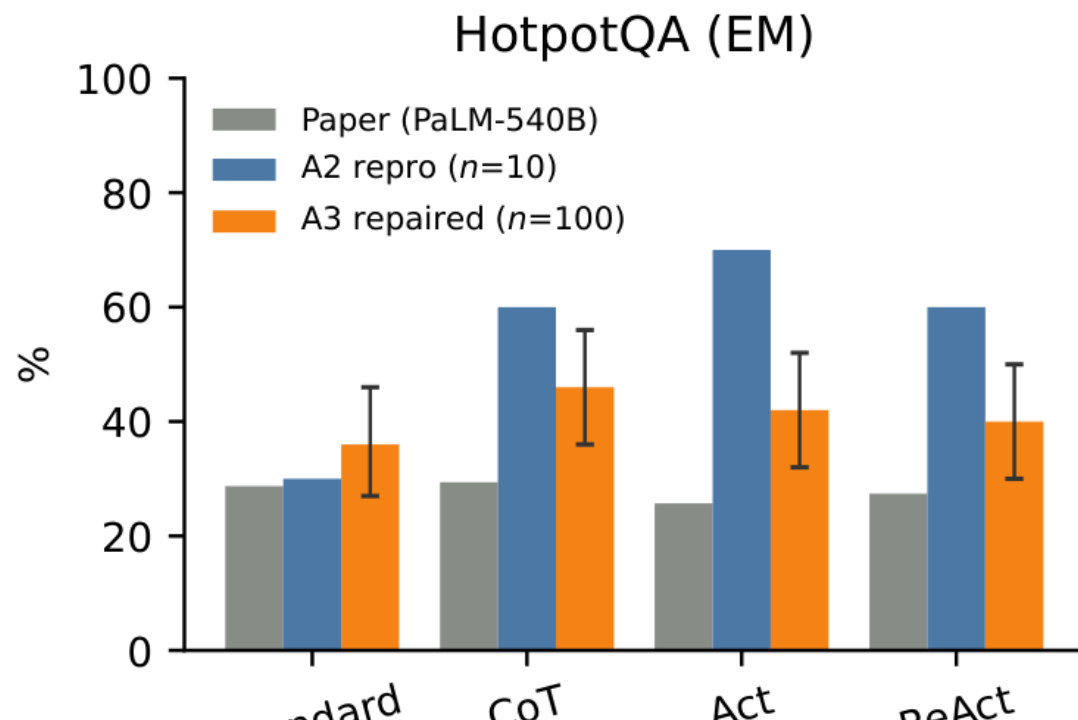
2domains

100items / domain

Two bugs fixed *before* touching the paper's idea

honesty check

- **Token truncation** — `max_tokens=256` cut completions off before `Answer: ;`; CoT-SC silently dropped empty votes. Raised to **512** for every arm equally.
- **No cost accounting** — added a thread-safe token/cost meter.



The limitation, found in the trajectories

read 20

real traces

8/10FEVER episodes: one Search, then Finish

0 times the paper's trigger fired on them

- Gold label `NOT ENOUGH INFO` → model predicted `REFUTES` , **6/6 times**
- HotpotQA: paper's trigger works fine — failures burn all 7 steps, no `Finish`
- **The trigger isn't wrong — it's incomplete.** It only sees "stuck," never sees "confidently wrong on thin evidence"

The fix: three signals

S1vS2vS3

= "CGA"

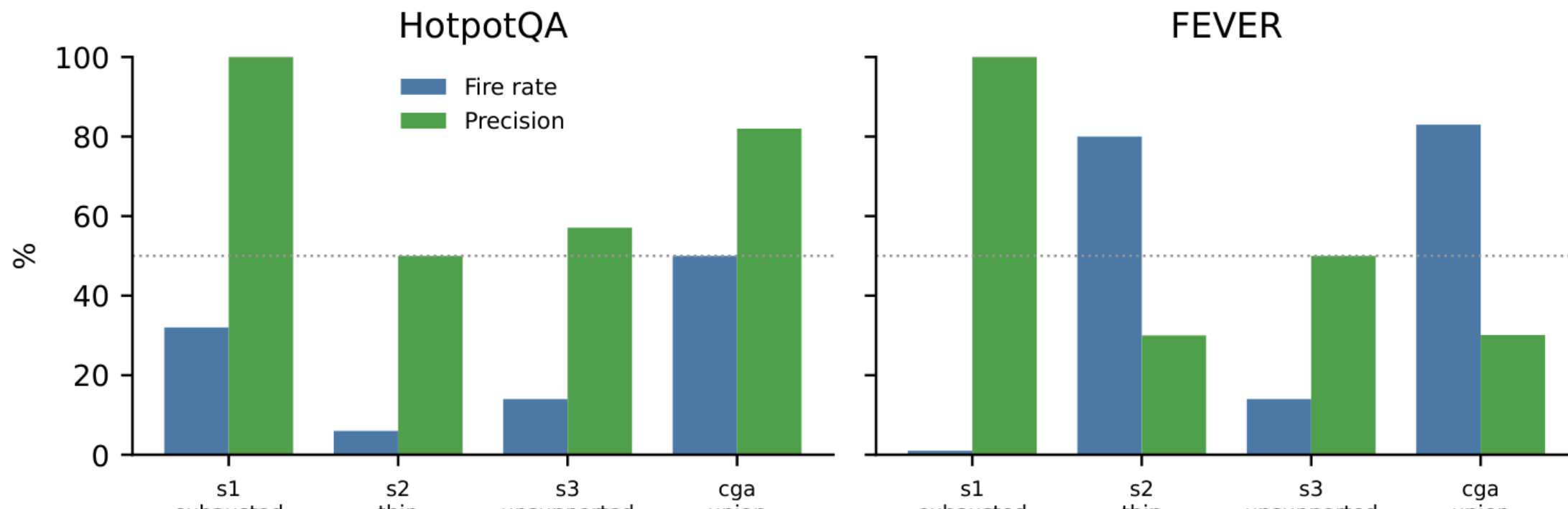
- S1 — step exhaustion (*paper's original condition, kept as control*)
- S2 — evidence thinness — fewer than τ informative observations before **Finish**
- S3 — entailment check — one extra LLM call: "*does the evidence support this answer?*" → back off on **INSUFFICIENT**

S3 costs 1 extra call, vs. the 21 calls of the CoT-SC arm it gates.

Validate before you pay

test cheap, then spend

- Score each signal by **precision lift** = $\text{precision} \div \text{base error rate}$
- A signal firing on *everything* trivially matches the base rate — contributes nothing
- Ran this on the existing ReAct traces — **before** committing to a full ablation



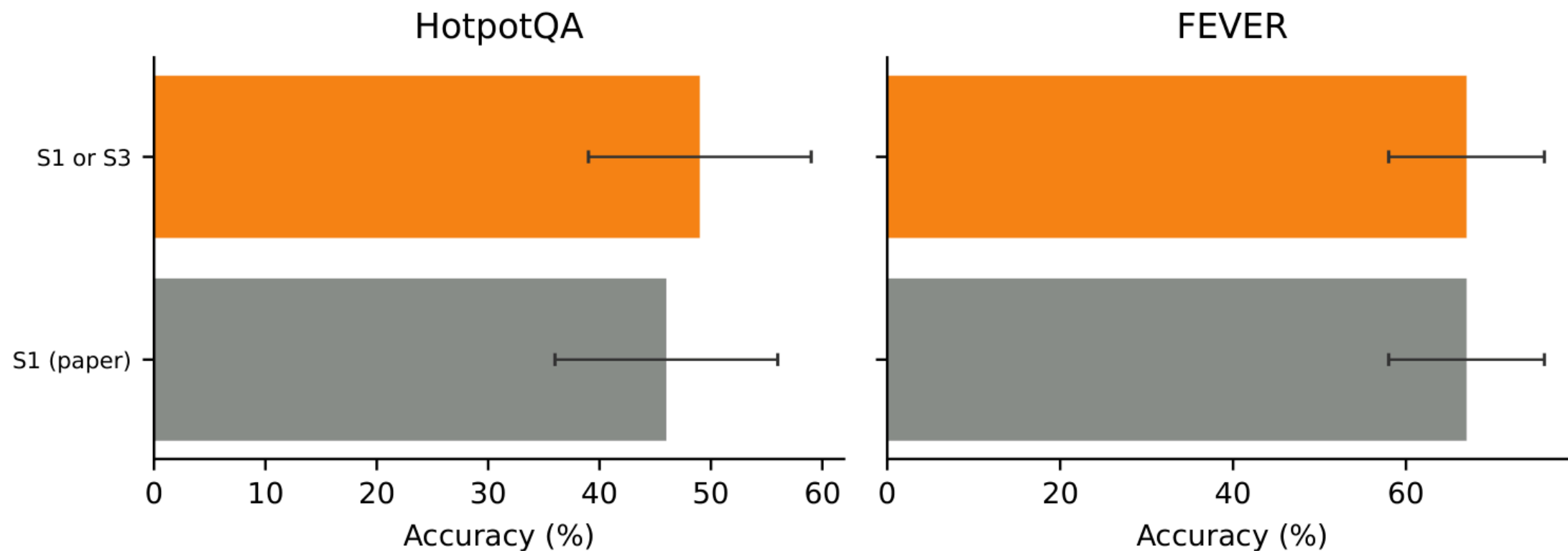
The negative result

reported, not
hidden

- **S2 (evidence count) does not work:**
 - FEVER: fires 80/100 times, only $1.07\times$ precision lift — near-degenerate
 - HotpotQA: $0.67\times$ lift — *below* the base rate
- Evidence *count* $\neq$ evidence *quality*
- **Decision:** dropped all S2-containing arms from the paid ablation, spent the freed budget on S1vS3 instead
- **S3 (entailment) works:** $1.79\times$ lift on FEVER; combined with S1, **86.4%** precision on HotpotQA

Ablation results

n=100,
paired McNemar

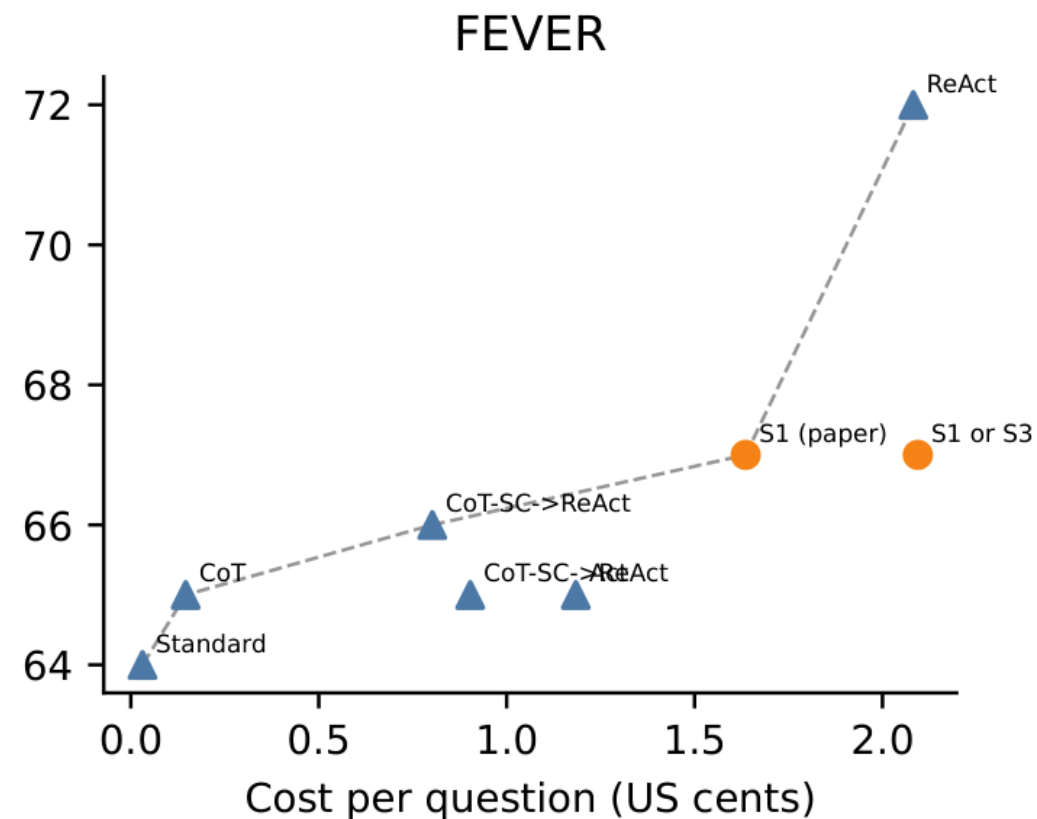
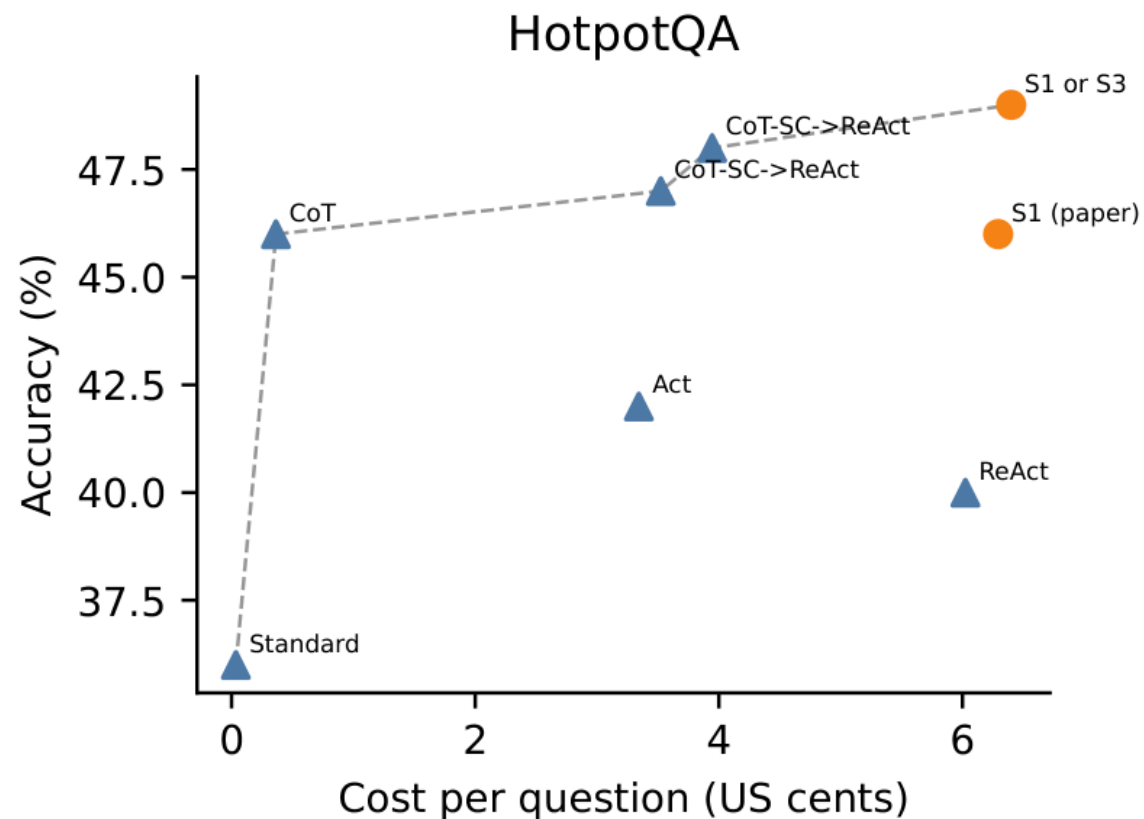


Domain	Direction	Arm	Acc %	Δpp

Cost vs. accuracy

+1 call

vs. 21 it gates



Total measured spend for the full study: **\$38.80** (trimmed from an initial 5×-too-low

Conclusion

the real
point

- Paper's trigger only sees "stuck" (step exhaustion) — it misses "confidently wrong on thin evidence," the failure mode that **dominates on FEVER**
- **S3 fixes that gap:** 1 extra LLM call — "*does the evidence support this answer?*" — catches what step-counting can't
- **The economics are the point:** that 1 call gates a CoT-SC fallback that costs **21 calls**. S3 decides *whether to pay 21×*, for the price of 1
- Net effect: same or better accuracy, without inflating cost on the ~85% of episodes that didn't need the expensive fallback anyway

thank you —
questions?